# KISII SCHOOL 2025 PREDICTION

## (KCSE PRE-ANALYSIS TEST)

**231/3**

# BIOLOGY

**PAPER 3 (PRACTICAL)**

**TIME: 1¾ HOURS**

**NAME……………………………………………………………………………………**

**SCHOOL………………………………………………… SIGN…………………………**

**INDEX NO……………………………………………… ADM NO……………………**

*(Kenya Certificate of Secondary Education)*

**INSTRUCTIONS TO CANDIDATE**

1. *Write your name and admission number in the spaces provided above.*
2. *Sign and write date of examination in spaces provided.*
3. *Answer all questions in the spaces provided in the question paper*
4. *You are not allowed to start working with the apparatus for the first 15 minutes of the 1 & ¾ hours allowed for this paper. This time is to enable you to read the question paper and make sure you have all the chemicals and apparatus that you may need.*

**For Examiners Use Only**

| Section | Question | Maximum score | Candidates score |
|---|---|---|---|
| | 1 | 15 | |
| | 2 | 15 | |
| | 3 | 10 | |
| TOTAL | | 40 | |

| SCORE | | |
|---|---|---|

**1. You** are provided with three solutions K, M3 & M4. M3 is the same as M4 except that M4 is boiled. You are also provided with 2 visking tubing, piece of thread, two beakers, 6 test tubes in a rack, thermometer, test tube holder and the following solutions; iodine, sodium hydroxide, copper(II) sulphate and Benedict's.

(a) Tie one end of the visking tubing tightly using a thread. Put 10ml of solution K and 1 ml of solution M3 into the visking tubing. Tie the other end of the visking tubing. Place the visking tubing in a beaker half filled with iodine solution. Take a second visking tubing and repeat the same using M4. Place the beakers containing the visking tubing in a water bath maintained between (35 – 38)$^0$C. Allow the set-ups to stand for 30minutes. Then after 30 minutes remove the visking tubings and test their contents and those of the beakers.

**SAMPLE 1 (K + M3) (6 MARKS)**

| CONTENTS IN | TEST | OBSERVATION | CONCLUSION |
|---|---|---|---|
| VISKING TUBING | Iodine test | | |
| | Benedict's test | | |
| | Biuret's test | | |
| BEAKER | Iodine test | | |

| | | | |
|---|---|---|---|
| | Benedict's test | | |
| | Biuret's test | | |

**<u>SAMPLE 2 (K + M4)</u> (6MARKS)**

| CONTENTS IN | TEST | OBSERVATION | CONCLUSION |
|---|---|---|---|
| VISKING TUBING | Iodine test | | |
| | Benedict's test | | |
| | Biuret's test | | |
| BEAKER | Iodine test | | |

| | | | |
|---|---|---|---|
| | Benedict’s test | | |
| | Biuret’s test | | |

**(b)**Account for the results in:

**(i)**Table 1 **(1mark)**

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**(ii)** Table 2 **(2marks)**

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**2. You** have been provided with specimen labelled X. The specimen and the photograph of Y are obtained from the same animal. Use them to answer the question that follow.

**PHOTOGRAPH Y**

**a)** Giving two reasons identify Y.

(i) Identity **(1mark)**

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**(ii)**Reasons **(2marks)**

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**b) Give two** adaptations of the specimen labelled X **(2marks)**

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**c) State two** observable differences between X and Y **(2marks)**

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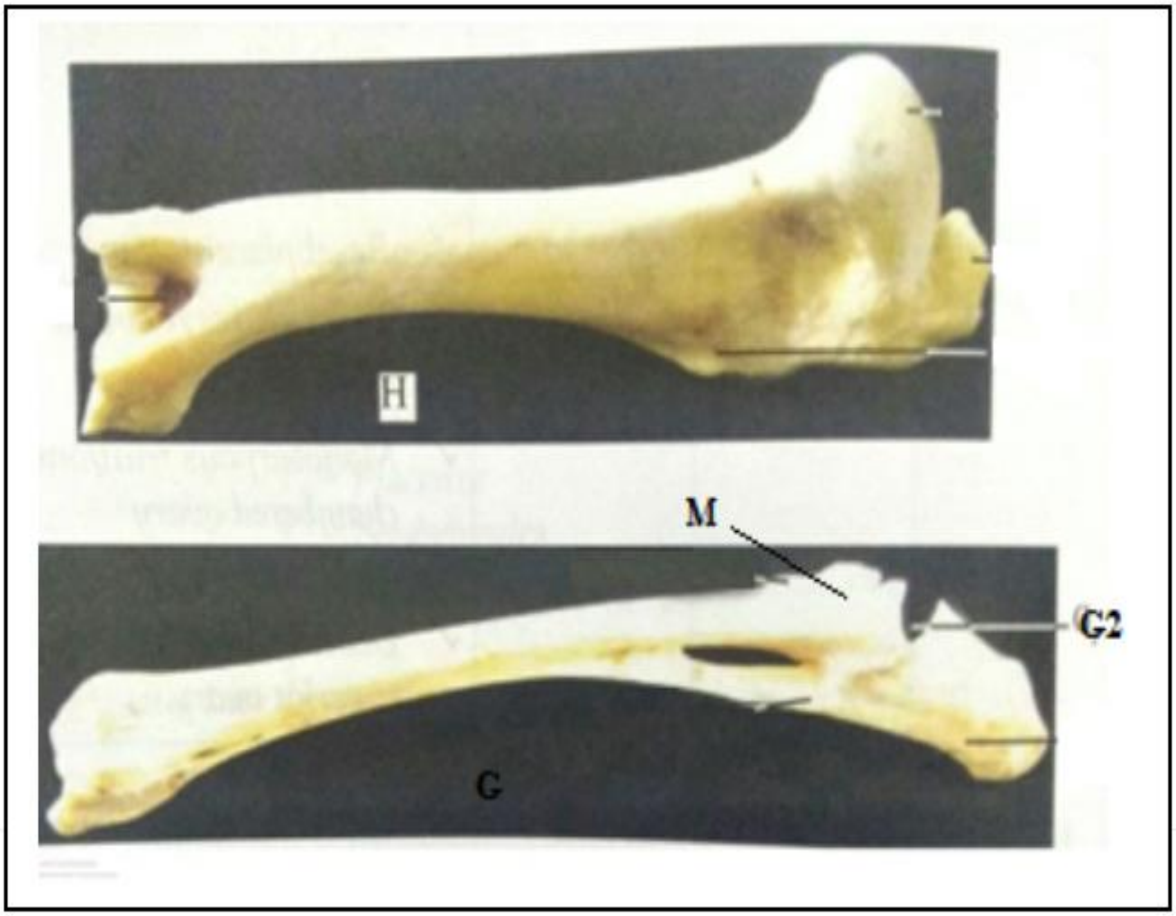
**d) Make** a drawing showing the anterior view of X and label all the parts **(4marks)**

**e) The** photograph below shows bones from another animal




**(i)** Which part of the body was H obtained from **(1mark)**

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**(ii)** Identify G2 **(1mark)**

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(**iii)** With a reason name the type of joint at the proximal end of specimen H **(2marks)**

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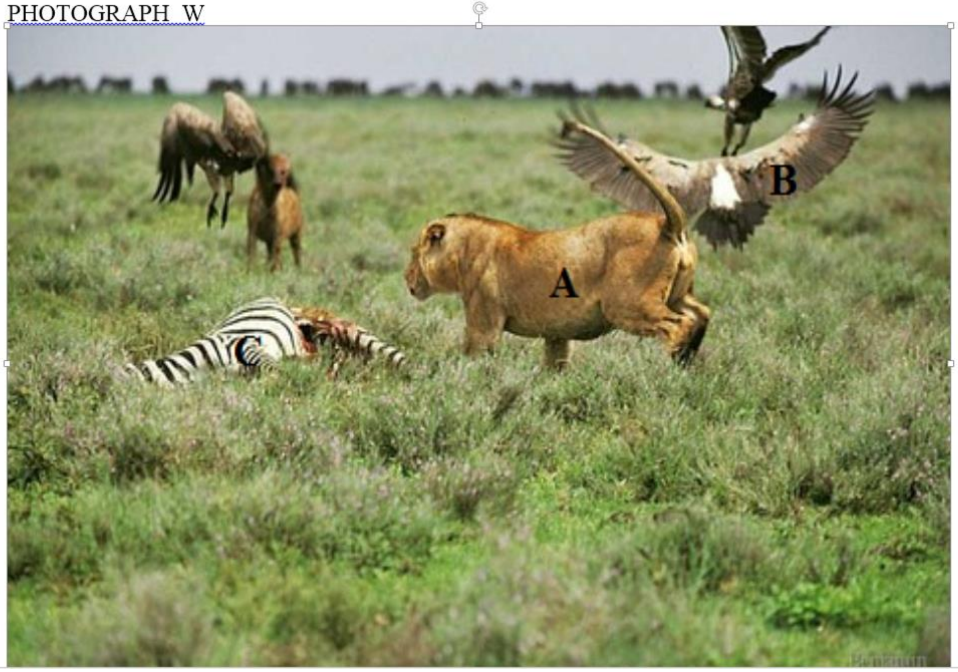
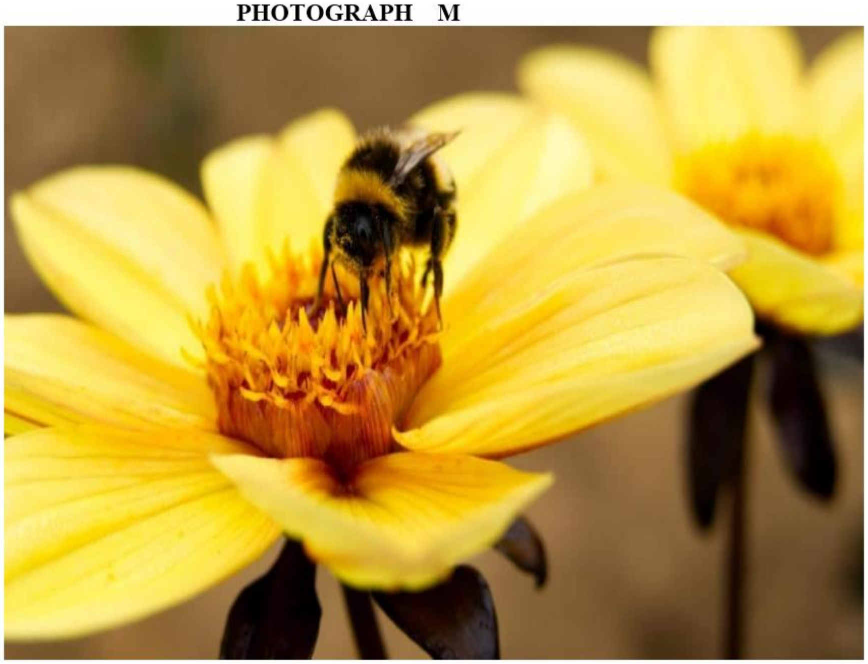
**3.** Study the photographs below and answer the questions that follow

PHOTOGRAPH W




**PHOTOGRAPH M**



**a)** Name the type of relationship in:

(i) Photograph W **(1mark)**

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**(ii)** Photograph M **(1mark)**

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**b) Explain** your answer in a(ii) above **(2marks)**

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**c)** What is the importance of the relationship taking place in the photograph W **(1mark)**

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**d) Using** observable feature only explain two ways in which the flower is adapted for the activity taking place in photograph M **(2marks)**

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**e).(i)**Give the biological term used to refer to the phenomenon captured in photograph W between organisms A and B towards C **(1mark)**

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**(ii)**State two implications of the phenomenon mentioned in d(i) above **(2marks)**

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